MILITARY SPECIFICATION

RECEIVER-TRANSMITTER, RADIO RT-547(*)/ASQ-19

AND

DECODER, PULSE KY-312(*)/ASQ-19

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide output signals to actuate panel instruments for display of distance and bearing to a TACAN navigation beacon.

1.2 Associated Equipments - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment.

Amplifier-Power Supply-Receiver AM-2349/ASQ-19

Earphone H-87A/U

Antenna AT-234/APX

Indicator, Bearing-Distance-Heading ID-663/U

Control, Central C-3076/ASQ-19

Intercommunication Station LS-460/AIC

Intercommunication Station LS-459/AIC

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-18309	Integrated Electronic Central AN/ASQ-19

STANDARDS

Military

MIL-STD-291	TACAN Standard Signal

FSC 5826

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Federal

FED-STD-595	Colors (collection of 358 standard colors for paints and other materials)

2.2 Availability of Documents:

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, parts and materials requirements shall be in accordance with Specification MIL-I-18309.

3.2 Design and Construction:

3.2.1 General - This equipment shall conform with all the applicable requirements of Specification MIL-I-18309 for design, construction, and workmanship, except in the paragraphs as modified below:

3.2.2 Total Weight - The total weight, excluding cables, shall not exceed 40.5 pounds for the RT-547/ASQ-19 and 30.5 pounds for the KY-312/ASQ-19.

3.2.4.1 Operational Stability - The equipment shall operate with optimum performance for 200 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.5.1 Cables and Connectors - Connectors used for interconnection of removable modular subassemblies shall be type DA manufactured by Cannon Electric Company, or type 143 manufactured by Amphenol Electronics Corporation, or approved equivalents. Printed card connectors may be used in modules for mating chassis connectors where applicable. External connectors shall be Amphenol Type 126-10753 and Amphenol Type 126-10572 or approved equivalents. The wires and terminals of this connector shall be potted with a sealing compound which is capable of withstanding temperatures to 125°C.

3.2.12 Standard Conditions - In addition to the requirements of Specification MIL-I-18309 the following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Cooling Air	(1) RT-547/ASQ-19 10 lbs/hr of ambient room air. (2) KY-312/ASQ-19 22 lbs/hr of ambient room air.
B+ and Bias Voltages	430 ±12 V DC, 250 ±7 V DC, 140 ±2 V DC, 130 ±4 V DC, -180 ±5 V DC

3.2.13.1 Minimum External Cooling Requirements - The equipment shall operate satisfactorily under the temperature-altitude conditions specified for Class 2 equipment per Specification MIL-E-5400 when external cooling air is supplied at the rate indicated in figure 1. The curve of figure 1 specifies the minimum cooling air flow rates that are required for air inlet temperature varying from -40°F to 120°F. The cooling air weight flow as a function of the RT-547/ASQ-19 and the KY-312/ASQ-19 equipment pressure drops is specified in the following equation.

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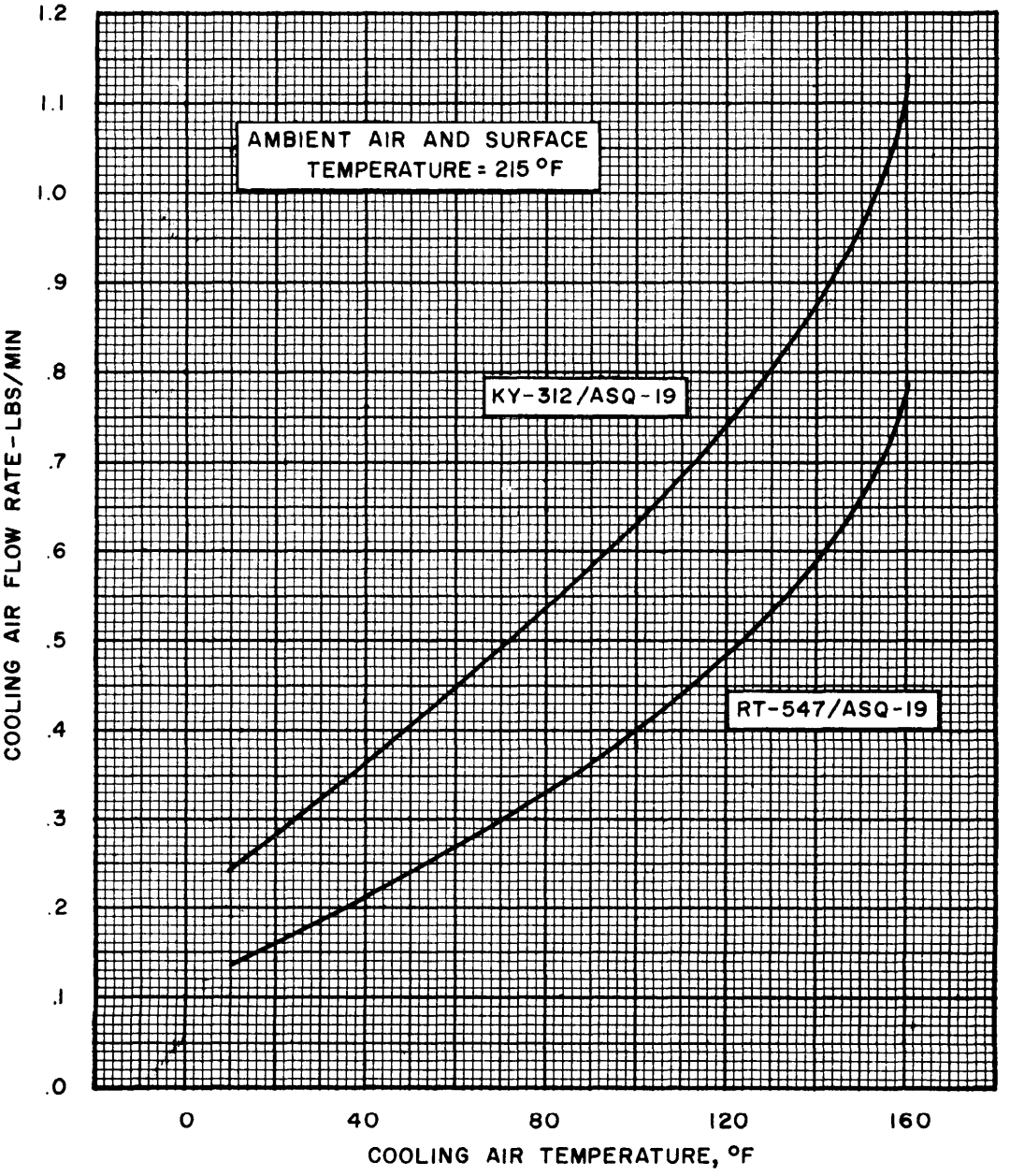


Figure 1. Cooling Air Flow Rate Vs Cooling Air Temperature

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RT-547/ASQ-19

$$W = \left[.373\ (\sigma\Delta\rho)^{0.66}\right] \pm 10\%\ \text{lb/min.}$$

KY-312/ASQ-19

$$W = \left[.578\ (\sigma\Delta\rho)^{0.53}\right] \pm 10\%\ \text{lb/min.}$$

Where

W = Cooling air flow rate (lb/min.)

$\sigma\Delta\rho$ = Corrected total pressure (inches H_2O)

σ = Density correction (ρ/ρ_r)

ρ_r = Standard density (0.0765 lb/ft^3)

3.3 Performance - Receiver-Transmitter, Radio RT-547/ASQ-19 and Decoder, Pulse KY-312/ASQ-19, when properly interconnected to an antenna, aircraft instruments and power source, shall provide the following functions:

(1) Determine the radial of the navigation beacon on which the aircraft is flying and actuate the bearing indicator to continuously display the magnetic bearing to the beacon.

(2) Determine the distance of the aircraft from the navigation beacon and actuate the distance indicator to continuously indicate the distance to the beacon.

(3) Determine the deviation of the aircraft from the selected course or beacon radial and actuate the vertical needle of the deviation indicator.

(4) Aural indication of the identity of the beacon to which the equipment is channeled.

(5) Facilities for the actuation of flag alarm circuits when distance and bearing indications are unreliable due to absence of proper input signal or failure in the equipment.

(6) Auxiliary functions for use with other equipment including blanking pulses synchronous with the transmitted pulses.

3.3.1 Electrical

3.3.1.1 Characteristics of TACAN Signals - This equipment shall operate with pulse code and pulse amplitude type signals of a ground facility meeting the requirements of MIL-STD-291.

3.3.1.2 Antenna Impedance - The equipment shall operate with antennas and coaxial r-f transmission lines of 52 ohms impedance. The input voltage standing wave ratio (VSWR) shall be not more than 4:1 on all channels. The equipment shall not be damaged by operation in the transmit-receive (T/R) mode with the antenna connector short-circuited or open-circuited.

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3.3.1.3 Output Requirements - The equipment shall provide suitable output circuits for operation with up to three ID-663/U indicator and one ID-387/ARN indicator with its deviation indicator terminals paralleled with a 500-ohm resistor.

3.3.1.4 Remote Control - Complete operational remote control of the radio receiver-transmitter shall be made through Control, Central C-3076/ASQ-19 or Control, Radio Set C-3146/ASQ.

3.3.1.5 Operational Range - The operational range of the equipment shall be such that reliable bearing, distance, and identity information will be maintained from zero out to a line of sight distance of 195 nautical miles from the beacon. Under no circumstances shall the equipment lock on its own transmission. No degradation of any of the other functions of the equipment shall occur as a result of its own transmission.

3.3.1.6 Distance Sensitivity and Accuracy - As measured at the output of the distance synchro transmitters with precision test equipment and dummy loads, the equipment shall have the following distance-measuring performance when operating with a beacon which is 90% efficient in replying to interrogations.

CONDITIONS	ACCURACY	SIGNAL LEVEL
Standard	± (0.2 miles +0.1%)	-117 dbw to -30 dbw
Service	± (0.3 miles +0.1%)	-114 dbw to -30 dbw

3.3.1.6.1 Distance Tracking Rate - When operating with a beacon which is 60% efficient in replying to interrogations, the equipment shall be capable of locking on and continuously tracking distance replies, while moving at speeds up to 2000 knots in either direction with respect to its own interrogation and with signal levels 6 db above the minimum signal necessary for lock-on.

3.3.1.7 Bearing Sensitivity and Accuracy - With nominal signal modulation, the bearing sensitivity and accuracy as measured with precision test equipment and dummy loads at the output of the fundamental synchro resolver and the synchro differential transmitter (with a simulated zero degree signal applied to the compass input) shall be as follows:

STANDARD CONDITIONS

BEARING	BEARING ACCURACY	SIGNAL LEVEL
270 Degrees	±2 Degrees	Not greater than -117 dbw
All Bearings	±2 Degrees	Not greater than -113 dbw
All Bearings	±1 Degree	From 6 db above minimum required for 90 degrees lock-on up to -30 dbw.

SERVICE CONDITIONS

BEARING	BEARING ACCURACY	SIGNAL LEVEL
270 Degrees	±2.5 Degrees	Not greater than -115 dbw
All Bearings	±2.5 Degrees	Not greater than -110 dbw
All Bearings	±1.5 Degrees	From 6 db above minimum required for 90 degrees lock-on up to -30 dbw

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3.3.1.7.1 Bearing Tracking Rate - The bearing system shall lock on and track signals having the bearing changing at rates up to 20 degrees per second in either direction for signal levels from 6 db above the minimum signal necessary for lock-on at 90 degrees bearing up to -30 dbw under standard and service conditions, except that at ambient temperatures below -30°C, the tracking rate shall be up to 10 degrees per second. After an additional 15 minutes of operation, the bearing system shall lock on and track signals having bearings changing at rates up to 20 degrees per second.

3.3.1.8 Automatic Gain Control - As the signal is varied from a level 6 decibels above the minimum required for lock-on at 90 degrees bearing up to -30 dbw, the accuracy shall remain within the limits specified in 3.3.1.6 and 3.3.1.7.

3.3.1.9 Coarse Bearing System Accuracy - The phase angle of the pulse produced from the 15 cps modulation with respect to the pulse produced from the north reference burst shall be held within ±7.5 degrees of the correct value under standard conditions and within ±10 degrees under service conditions. This requirement shall be met with signal modulation levels from 10 to 55%.

3.3.1.10 Beacon Identification - The equipment shall provide intelligible and unambiguous identification output with signal inputs from -117 dbw to -30 dbw under standard conditions, and -114 dbw to -30 dbw under service conditions. Audio output shall be a 1350-cps tone of not less than 50 milliwatts nor more than 500 milliwatts as measured with a 600-ohm load. Signal-to-noise ratio (measured with tone and squitter signals applied in sequence) shall be not less than 30 db at signal levels 3 db or more above the minimum levels specified above. Accidental shorting or opening of the external load shall not cause damage to the equipment or render other functions inoperative. The output circuit shall be ungrounded.

3.3.1.11 Selectivity

3.3.1.11.1 Adjacent and Other Channels - The equipment shall have 50 db or greater rejection to any normal beacon signal or signals on channels other than the channel to which the equipment is tuned. This requirement applies both in the presence or absence of a desired beacon signal on the selected frequency channel. In the presence of a desired signal, the undesired signal shall be considered rejected if the performance requirements of this specification are met, with the exception of allowing an additional 1-degree bearing error. In the absence of a desired signal, the undesired signal shall be considered rejected if no distance, bearing, or identity tone information is obtained.

3.3.1.11.2 Co-Channel - With nominal signal modulation, the equipment shall be capable of rejecting the weaker of two beacon signals received on the same channel when the difference in signal strength is 8 db or more. The performance obtained from the stronger signal shall meet the requirements of this specification, except that an additional 1-degree bearing error shall be permitted when the two signals differ in level by 8 db.

3.3.1.12 Decoding - The sensitivity of the receiver shall not decrease more than 3 db when receiving pulse pairs spaced ±0.5 microsecond from nominal, and loss of receiver sensitivity shall be at least 30 db for pulse pair spacing ±4.0 microseconds or greater from nominal.

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3.3.1.13 Frequency Channeling - 126 crystal-controlled frequency channels shall be provided at 1.0 mc spacing. Channels 1 through 63 shall pair receiver frequencies 962 through 1024 mc with transmitter frequencies 1025 through 1087 mc; and channels 64 through 126 shall pair receiver frequencies 1151 through 1213 mc with transmitter frequencies 1088 through 1150 mc.

3.3.1.14 Channel Selection Time - The time required by the equipment to tune from any channel to any other channel shall not exceed 9 seconds.

3.3.1.15 Calibration - Under standard conditions the equipment shall be adjusted to read zero miles ± 0.1 mile as measured at the output of the units synchro transmitter when the delay between the 50% amplitude points of the second r-f interrogation pulse and the second r-f reply pulse is 50 microseconds.

3.3.1.16 Peak Power Output - The peak power of each pulse in a pulse pair transmitted by the equipment in the T/R mode of operation during search conditions shall be as follows:

Standard Conditions	+30 dbw min.
Service Conditions	+27 dbw min.

The two pulses in a pulse pair shall not differ in power level by more than 1 db. The above power measurement shall be determined using a 52-ohm load impedance. Not less than 55% of full transmitter output power shall be delivered to a load presenting a standing wave ratio of 2:1 at any phase angle.

3.3.1.17 Modulation Characteristics - The output pulse envelope as detected by a linear detector shall have a rounded shape falling within the limits outlined below.

3.3.1.17.1 Pulse Shape - The pulse envelope as detected by a linear detector shall have a rounded shape falling within the following limits:

3.3.1.17.2 Pulse Top - The amplitude of each pulse between the 90% amplitude points shall be flat to within ± 1 db.

3.3.1.17.3 Pulse Duration - The pulse duration measured between 50% amplitude points shall be 3.5 ± 0.5 microseconds.

3.3.1.17.4 Pulse Coding - The pulse code spacing measured between 50% amplitude points from leading edge to leading edge shall be 12 ± 0.5 microseconds.

3.3.1.17.5 Pulse Repetition Rate - The pulse repetition rate shall fall between 120 and 150 pulse pairs per second in the track condition. The pulse repetition rate shall vary or jitter enough to preclude synchronization of transmitted pulses between any of 100 equipments on the same channel for more than one second in one hour.

3.3.1.18 Transmitter Spectrum - In both the search and track conditions, 90% of the transmitted energy shall be within a 500 kc band centered on the nominal channel frequency of that channel, and the remaining energy shall decay smoothly with frequency.

3.3.1.19 Speed of Response

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3.3.1.19.1 Distance Search Time - A complete distance search cycle shall not exceed 30 seconds, with or without a signal applied.

3.3.1.19.2 Bearing Search Time - A complete bearing search cycle shall not exceed 18 seconds with or without a signal applied except that, at ambient temperatures below -30°C, a complete bearing search cycle shall not exceed 40 seconds after the initial 5 minutes warmup time. With an additional 15 minutes of operation, a complete bearing search cycle shall not exceed 18 seconds.

3.3.1.19.3 Distance Memory - Upon loss of signal, the memory period during which the equipment retains its distance reading shall fall between 4 and 8 seconds.

3.3.1.19.4 Bearing Memory - Upon loss of signal, the memory period during which the equipment retains its bearing reading shall fall between 8 and 12 seconds.

3.3.1.20 Stability

3.3.1.20.1 Transmitter and Local Oscillator Stability - Each of the radio receiver-transmitter frequency channels shall be maintained within ±100 kc of its assigned frequency. Up to a 2:1 standing wave ratio at the antenna terminal shall not cause the transmitter frequency to fall outside the specified limits.

3.3.1.20.2 I-F Stability - The i-f center frequency of the receiver-transmitter shall be maintained within ±50 kc of 63 mc from minus 20°C to plus 85°C. At minus 54°C the i-f center frequency shall be maintained within ± 140 kc of 63 mc.

3.3.1.21 Instrumentation Output Characteristics

3.3.1.21.1 Course Deviation - With three standard 1000-ohm deviation indicator loads (333-ohm load), the course deviation output circuit shall be adjustable to a sensitivity of 5 degrees course error equal to a full scale indication of 150 microamperes per indicator (150 microamperes in each 100-ohm load.) This is a 10 degrees total course width pin to pin.

3.3.1.21.2 To-From Circuit - The to-from circuit shall be capable of supplying the ambiguity indicator of Indicator, Course ID-387/ARN with d-c either polarity and a minimum magnitude of 250 microamperes for ±20 degrees error. Polarity of current shall be determined by the relative phase of the 400-cps signal input to the to-from circuit.

3.3.1.21.3 Bearing

3.3.1.21.3.1 Relative Bearing - A synchro of the differential control type shall be provided to combine an electrical compass heading signal from an Eclipse Pioneer AY-201-1 or equivalent synchro transmitter so as to produce an electrical output suitable for actuating the synchro receiver-driven relative bearing pointers of the Indicator, Bearing-Distance-Heading ID-663/U.

3.3.1.21.3.2 Magnetic Bearing - A synchro resolver rated for 26-V 400-cps excitation and 12.6-volt line-to-line maximum output shall be provided for magnetic bearing electrical output. The resolver shall be set at 400 cps electrical zero (EZ) at 300 degrees indicated bearing.

3.3.1.21.3.3 Harmonic Bearing Resolver - Provisions shall be included for the addition of a harmonic bearing resolver and gear train. The space allotted for the addition of the harmonic bearing resolver and gear train shall be consistent with harmonic bearing resolver operation at nine times fundamental bearing speed.

3.3.1.21.4 Distance

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3.3.1.21.4.1 Synchro Transmitters - Synchro transmitters, rated at 26-volt 400-cps input, shall be provided to produce electrical distance output signals to the synchro torque receivers driving the units and tens dials of the ID-663/U.

3.3.1.21.4.2 Hundreds Dial Switch - A switch shall be provided, actuated by the distance gear train, for the purpose of controlling the hundreds dial of the ID-663/U. When the measured distance is between 0 and 99 miles, 400-cps positioning voltages shall be supplied so as to cause a zero indication of the hundreds dial of the ID-663/U. When the measured distance is between 100 and 199 miles, the switch shall be actuated so as to cause an indication of one on the hundreds dial of the ID-663/U.

3.3.1.21.4.3 Distance Takeoff Potentiometer - Dual potentiometers with ratings of 3 watts nominal and 2 watts at 80°C, 2000 ohms ±1%, linearity ±0.15% for 350 degrees, with 360 degrees of rotation and a dead angle of 5.5 degrees or less shall be provided to develop output voltage proportional to distance. The distance takeoff potentiometers shall be ganged with the distance synchro transmitters.

3.3.1.22 Flag Alarm Circuits

3.3.1.22.1 Bearing Flag Alarm - The bearing flag alarm circuit shall provide a terminal which is grounded when the bearing circuits are properly tracking a TACAN signal, and which is ungrounded when the bearing circuits are not properly tracking.

3.3.1.22.2 Distance Flag Alarm - The distance flag alarm circuit shall provide a terminal which is grounded when the distance circuits are properly tracking a TACAN signal, and which is ungrounded when the distance circuits are not properly tracking.

3.3.1.23 Auxiliary Functions

3.3.1.23.1 Blanking Pulse Output - The equipment shall provide a blanking pulse output with the following characteristics:

(1) Positive polarity.

(2) 20 to 70 volts amplitude across 1000 ohms in parallel with a 500 micro-microfarad load impedance.

(3) 20 volts per microsecond minimum rise time.

(4) One to five microsecond duration.

(5) 10 volts per microsecond minimum decay (90% to 10% amplitude).

(6) One pulse shall be generated for each interrogation out of the equipment and the leading edge of this pulse shall occur at least 0.8 but not more than 5 microseconds before the 10% amplitude point of the leading edge of the first pulse of the interrogation pulse pair.

(7) Spurious output associated with the blanking pulse shall be below 1 volt in amplitude.

(8) A blanking pulse shall be generated only when the function is in the transmit condition of operation.

3.3.1.23.2 Automatic Antenna Selector Output - A suitable audio signal shall be available from the equipment for operation of an automatic antenna selector. The presence of this signal shall signify that the proper signal type and level is received at the antenna terminal for operation of the equipment. The output shall have the following characteristics:

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(1) Amplitude between 0.5 and 20 volts rms into a load having a resistance not less than 5,600 ohms.

(2) Any frequency between 100 and 10,000 cps.

3.3.1.24 Power Requirements - The equipment shall give specified performance with the power sources listed in Specification MIL-I-18309. The power requirements shall not exceed the specified amounts.

(1) A-C Power (Single Phase)

26-Volts A-C

Current	1.5 amperes

115-Volts A-C

Current	2.2 amperes

(2) D-C Power 28-Volt System

Current, Channeling	8 amperes

(3) D-C Power (From AM-2349/ASQ-19)

(a) +430 V

Operating Voltage Limits	378 to 495 volts
Current	0.006 at 430 volts

(b) +250 V

Operating Voltage Limits	221 to 274 volts
Current	0.034 ampere at 250 volts

(c) +140 V

Operating Voltage Limits	137 to 143 volts
Current	0.220 ampere at 140 volts

(d) +130 V

Operating Voltage Limits	117 to 140 volts
Current	0.094 ampere at 130 volts

(e) -180 V

Operating Voltage Limits	-152 to -203 volts
Current	0.018 ampere at -180 volts

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3.4 Detail Requirements

3.4.1 Receiver-Transmitter, Radio RT-547/ASQ-19 - The radio receiver-transmitter shall meet the following requirements.

3.4.1.1 Function - When properly installed in an AN/ASQ-19, the radio receiver-transmitter shall perform the following functions:

(1) In response to trigger signals supplied from Decoder, Pulse KY-312/ASQ-19, generate pulsed r-f signals for interrogation of the TACAN navigation beacon on the selected frequency channel.

(2) Receive and demodulate the beacon bearing signals, distance replies, and identification signals.

(3) Provide a video output signal containing the composite demodulated bearing signals, distance replies and identification signals at a level suitable for operation of Decoder, Pulse KY-312/ASQ-19.

3.4.1.2 Form Factor - The radio receiver-transmitter shall be contained in a single unit capable of being directly mounted to a shockmounted equipment shelf which in turn is mounted in an aircraft. The over-all dimensions shall not exceed 7-17/32 by 7-33/64 by 22-37/64 inches. The front panel shall extend an additional 1-1/32 inches below the base.

3.4.1.3 Contents - The radio receiver-transmitter shall include a multifrequency crystal-controlled receiver and transmitter, modulator, and other circuitry required for performance of the specified functions. The unit shall be contained in a pressurized case and shall include a heat exchanger and internal blower as necessary for cooling. The following major subassemblies are to be included.

R-f module

Modulator module

Base chassis, with internal blower, heat exchanger, wiring and converters.

3.4.1.4 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
2A3J18	Industrial Products 81450 (or approved equivalent)	Antenna Connector
2A3J1	Amphenol 126-10753 (or approved equivalent)	Power and Control Connections

3.4.1.5 External Finish - The equipment covered by this specification shall be painted with color No. 2615 of FED-STD-595. The bottom surface of the equipment shall not be painted.

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3.4.2 Decoder, Pulse, KY-312/ASQ-19 - The pulse decoder shall meet the following requirements.

3.4.2.1 Function - When properly installed in the AN/ASQ-19, the pulse decoder shall perform the following functions:

(1) Supply to Receiver-Transmitter, Radio RT-547/ASQ-19 trigger pulses for controlling the r-f interrogation pulses.

(2) Accept the composite video signal from Receiver-Transmitter, Radio RT-547/ASQ-19 and utilize this signal to determine distance and bearing relative to the TACAN navigation beacon.

(3) Provide an audio frequency output signal suitable for aural identification of the TACAN navigation beacon.

(4) Provide a synchro resolver output which will supply Indicator, Course ID-387/ARN magnetic bearing information.

(5) Provide suitable output signals for the operation of the deviation indicator and to-from indicator of Indicator, Course ID-387/ARN.

(6) When supplied a magnetic heading signal from the aircraft compass system, provide a synchro output suitable for actuating a bearing pointer of the ID-663/U.

(7) Provide facilities for the actuation of flag alarm circuits when distance and bearing indicators are unreliable due to absence of proper signal input or failure in the equipment.

(8) Supply synchro signals for actuating the distance dials of the ID-663/U.

(9) Provide a dual linear distance potentiometer with terminals available for use with external equipment.

(10) Make provision for the addition of a harmonic bearing resolver, driving gears and associated wiring.

(11) Provide a blanking pulse output suitably synchronized with transmitted interrogations for suppression of other equipment during transmit periods.

(12) Provide an automatic antenna selector output consisting of an audio-frequency signal signifying that an adequate signal is being received for satisfactory radio set operation.

3.4.2.2 Form Factor - The pulse decoder shall be contained in a single unit capable of being directly mounted to a shockmounted equipment shelf which is in turn mounted in an aircraft. The over-all dimensions shall not exceed 7-9/16 (*) by 6-15/32 by 22-33/64 inches.

* Front plate extends an additional 1-1/32 inches below base of unit.
All dimensions are given in inches.

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3.4.2.3 Contents of Pulse Decoder KY-312/ASQ-19 - The pulse decoder unit shall contain all circuitry and mechanisms necessary to perform the required functions. The following major subassemblies are to be included:

Interrogation Generator Module

Distance Servo Amplifier Module

Time Discriminator Module

Distance Decoder Module

Fine Bearing Detector Module

Coarse Bearing Detector Module

Bearing Decoder Module

Bearing Servo Amplifier Module

Bearing Mechanism Module

Distance Mechanism Module

Phase Detector Module

Relay Module

Chassis, with connectors and wiring

Cover

3.4.2.4 Controls

Identification	Location	Function
3A6R3	Fine Bearing Detector Module	Track Adjust
3A6R9	Fine Bearing Detector Module	Zero Adjust
3A7R10	Coarse Bearing Detector Module	15 cps Gate Centering Adjust
3A9R47	Distance Decoder Module	Video Level Adjust
3A10R14	Time Discriminator Module	Gate Adjust
3A12R7	Interrogation Generator Module	Oscillator Level
3A12R13	Interrogation Generator Module	Zero Adjust
3A12R24	Interrogation Generator Module	Resolver Phase Adjust
3A12R48	Interrogation Generator Module	PRF Adjust
3A3R9	Phase Detector Module	Error Adjust
3A3R10	Phase Detector Module	Error Adjust

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3.4.2.5 Electrical Connectors - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
3A13J1	Amphenol 126-10572 (or an approved equivalent)	Power and Control Connections

3.4.2.6 External Finish - The equipment covered by this specification shall be painted with color no. 2615 of FED-STD-595. The bottom surface of the equipment shall not be painted.

3.4.2.7 Test Panel - Test points shall be provided on the front of the KY-312/ASQ-19 for use with suitable test equipment. The following voltages shall be provided to the test points.

Function	Test Point Designation
VIDEO	J22
AGC	J23
26 V AC	J24
28 V DC	J25
430 V DC	J26
250 V DC	J27
140 V DC	J28
130 V DC	J29
-40 V DC	J30
FIL	J31
MOD TRIG	J32
DIST. REPLY	J33
ANT. SW	J34

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Preparation for delivery shall be in accordance with Specification MIL-I-18309 to determine compliance with all applicable requirements.

5. PREPARATION FOR DELIVERY

5.1 General - Preparation for delivery shall be in accordance with Specification MIL-I-18309.

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6. NOTES

6.1 Use - The equipment covered by this specification is intended for aircraft installation as a portion of the Integrated Electronic Central AN/ASQ-19. When properly installed and operated the equipment provides output signals to actuate panel instruments for display of distance and bearing to the navigation beacon, deviation from selected course, and flag alarms. An aural signal is provided for identification of the navigation beacon.

6.2 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) System Specification - Integrated Electronic Central AN/ASQ-19 Specification MIL-I-18309 shall have precedence over this specification.

(3) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the procuring activity.

(4) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.10 of Specification MIL-I-18309. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purposes other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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